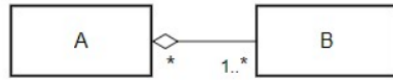


# Teste formativo 2020

## Pergunta 1

Por responder Pontuação 0,8 Destacar pergunta

Select the correct answer(s) about the given diagram:



Selecione uma ou mais opções de resposta:

- ☐ a. If an instance of A is deleted, the instances of B it contains are also deleted
- ☐ b. B is part of A
- ☐ c. The same instance of B can only be part of one instance of A
- ☐ d. An instance of B can exist without the existence of any instance of A

## Pergunta 2

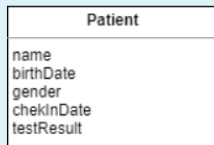
Por responder Pontuação 0,8 Destacar pergunta

In an UML class diagram, atomic attributes are those which:

Selecione uma opção de resposta:

- ☐ a. Can identify an object uniquely
- ☐ b. I don't want to answer
- ☐ c. Have a domain containing only atomic values and the attribute contains only a single value from that domain
- ☐ d. Can be determined from other attribute(s)
- ☐ e. Are part of a key

Consider the conceptual model of a hospital's database, where information on patients admitted with COVID-19 is stored. It is intended to change the model, represented below, to make it possible to store the evolution of the patients' situation regarding the infection (infected, recovered, reinfected). What is the best way to proceed?



Selecione uma opção de resposta:

- ☐ a. 

| Patient        |
|----------------|
| name           |
| birthDate      |
| gender         |
| checkInDate    |
| lastTestResult |
| numOfTests     |
- ☐ b. 

| Patient   | Test   |
|-----------|--------|
| name      | 1      |
| birthDate | 0..*   |
| gender    | date   |
|           | result |
- ☐ c. 

| Patient   | Test   |
|-----------|--------|
| name      | 0..*   |
| birthDate | 1      |
| gender    | date   |
|           | result |
- ☐ d. I don't want to answer
- ☐ e. 

| Patient     |
|-------------|
| name        |
| birthDate   |
| gender      |
| checkInDate |
| isInfected  |
| isRecovered |

## Pergunta 4

Por responder Pontuação 0,8 Destacar pergunta

When should we use a generalization?

Selecione uma opção de resposta:

- ☐ a. When subclass is the whole and the superclass is the part
- ☐ b. I don't want to answer
- ☐ c. When both super and subclass have attributes in common
- ☐ d. When subclass is a member of superclass
- ☐ e. When subclass is the part and the superclass is the whole

## Pergunta 5

Por responder Pontuação 0,8 Destacar pergunta

In UML class diagram, how does a derived attribute affect redundancy?

Selecione uma opção de resposta:

- ☐ a. Stored derived attributes are always redundant
- ☐ b. Stored derived attributes are not redundant
- ☐ c. Due to the high possibility of gaining additional knowledge from a derived attribute, they are always redundant
- ☐ d. I don't want to answer
- ☐ e. Since you store the derived attribute and all data necessary to derive it, derived attributes cannot affect redundancy.

## Pergunta 6

Por responder Pontuação 0,8 Destacar pergunta

When we want to convert many-to-one associations to the relational model by adding an additional relation with a key from the many side, what advantages will it bring to the schema?

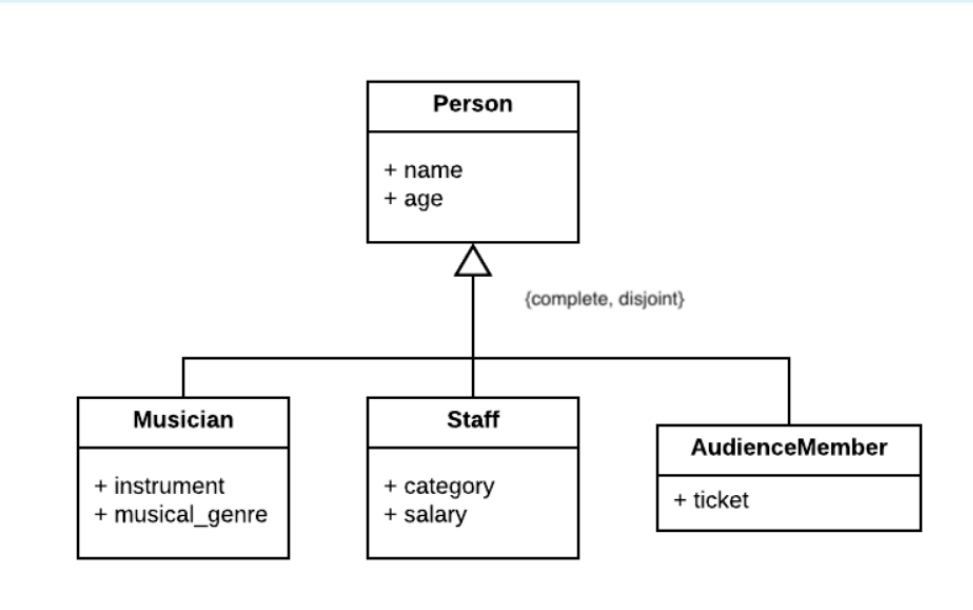
Selecione uma ou mais opções de resposta:

- ☐ a. Increased performance due to a smaller number of relations
- ☐ b. Increased performance due to a larger number of relations
- ☐ c. Increased rigour of the schema
- ☐ d. Increased extensibility

## Pergunta 7

Por responder Pontuação 0,8 Destacar pergunta

Consider the following UML diagram and its conversion to the relational model. Which are the two most adequate conversions?



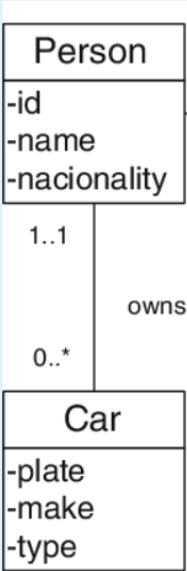
Selecione uma ou mais opções de resposta:

- ☐ a. Musician(id, name, age, instrument, musical\_genre)  
Staff(id, name, age, category, salary)  
AudienceMember(id, name, age, ticket)
- ☐ b. Musician(name, instrument, musical\_genre)  
Staff(name, category, salary)  
AudienceMember(name, ticket)
- ☐ c. Person(id, name, age, instrument, musical\_genre, category, salary, ticket)
- ☐ d. Person(id, name, age)  
Musician(id→Person, name, instrument, musical\_genre)  
Staff(id→Person, name, category, salary)  
AudienceMember(id→Person, name, ticket)

Pergunta 8

Por responder Pontuação 0,8 Destacar pergunta

Consider the following conceptual model. Mapping it to the relational model



Selecione uma ou mais opções de resposta:

- ☐ a. Can result in 2 relations with a foreign key in Car
- ☐ b. Can result in 3 relations, one of them with a compound primary key
- ☐ c. Can result in 3 relations, two of them without foreign keys
- ☐ d. Can result in 2 relations with a foreign key in Person

Pergunta 9

Por responder Pontuação 0,8 Destacar pergunta

There are three main methods that convert an UML into a Relation regarding Generalizations. Which one(s) is/are commonly used for overlapping generalizations?

Selecione uma ou mais opções de resposta:

- ☐ a. Object-oriented
- ☐ b. Use of NULLs
- ☐ c. E/R style

## Pergunta 10

Por responder

Pontuação 0,8

🚩 Destacar pergunta

Mapping a ternary association and its associated classes

Selecione uma ou mais opções de resposta:

- ☐ a. Is affected by the multiplicity
- ☐ b. Always results in 4 relations
- ☐ c. Is affected by the number of associated classes
- ☐ d. Can be converted to 3 relations

## Pergunta 11

Por responder

Pontuação 0,8

🚩 Destacar pergunta

A table in Boyce Codd Normal Form:

Selecione uma ou mais opções de resposta:

- ☐ a. Can have redundancy in some cases.
- ☐ b. May be subject of update anomalies.
- ☐ c. Cannot be subject of deletion anomalies.
- ☐ d. Cannot assure dependency preservation.

## Pergunta 12

Por responder

Pontuação 0,8

🚩 Destacar pergunta

Consider the relation R (A, B, C, D, E, F, G, H, I) with the following functional dependencies:

A -> D;

B -> I;

B, D -> H;

A, E -> C;

F, G -> E.

Which attributes belong to the closure of {A, F, G}?

Selecione uma ou mais opções de resposta:

- ☐ a. A
- ☐ b. B
- ☐ c. C
- ☐ d. D
- ☐ e. E
- ☐ f. F
- ☐ g. G
- ☐ h. H
- ☐ i. I

## Pergunta 13

Por responder

Pontuação 0,8

🚩 Destacar pergunta

Regarding decompositions, identify the correct statements.

Selecione uma ou mais opções de resposta:

- ☐ a. If we decompose to BCNF, the final schema will also be in 3NF
- ☐ b. If  $R_1 \bowtie R_2 = R$ ,  $R_1$  and  $R_2$  make a decomposition of  $R$
- ☐ c. If  $R_1$  and  $R_2$  are a decomposition of  $R$ , the functional dependencies of  $R$  are always preserved in  $R_1$  or  $R_2$
- ☐ d. If  $R_1$  and  $R_2$  are a decomposition of  $R$ , the functional dependencies of  $R$  are preserved in  $R_1$  and  $R_2$

# Pergunta 14

Por responder Pontuação 0,8 Destacar pergunta

Consider the relation R (A, B, C, D, E, F, G, H, I) with the following functional dependencies:

- A -> D;
- B -> I;
- B, D -> H;
- A, E -> C;
- F, G -> E.

What are the functional dependencies of the relation S (A, B, H), a projection of R in 3 attributes?

Selecione uma opção de resposta:

- ☐ a. A, B -> H
- ☐ b. A->D; B->I; B, D-> H; A, E->C
- ☐ c. I don't want to answer.
- ☐ d. A->D; B->I; A, B -> D, H, I; A, H->D; B, H->I
- ☐ e. S has the same functional dependencies as R

# Pergunta 15

Por responder Pontuação 0,8 Destacar pergunta

What is a minimal superkey?

Selecione uma ou mais opções de resposta:

- ☐ a. Candidate key
- ☐ b. Foreign key
- ☐ c. Primary key
- ☐ d. Secondary key

# Pergunta 16

Por responder Pontuação 0,8 Destacar pergunta

Consider you select a single numeric attribute as the primary key of a table. What is the most suitable data type for the primary key column in SQLITE?

Selecione uma opção de resposta:

- ☐ a. INTEGER if autoincrement is needed and REAL otherwise.
- ☐ b. INTEGER because SQLite will automatically create an implicit column called rowid.
- ☐ c. INTEGER OR REAL depending on the characteristics of the value.
- ☐ d. REAL if a big range of values is needed and INTEGER otherwise.
- ☐ e. I don't want to answer.

# Pergunta 17

Por responder Pontuação 0,8 Destacar pergunta

We wish to add a foreign key to the table T that references table A (the primary key of A is an int called AID).  
We also wish to update the foreign key when the primary key of A is updated.

Selecione uma opção de resposta:

- ☐ a. ALTER TABLE T ADD AID int REFERENCES A(AID) ON UPDATE CASCADE;
- ☐ b. ALTER TABLE T ADD AID int REFERENCES A(AID) ON UPDATE PROPAGATE;
- ☐ c. MODIFY TABLE T ADD AID int REFERENCES A(AID) ON UPDATE CASCADE;
- ☐ d. I don't want to answer
- ☐ e. MODIFY TABLE T ADD AID int REFERENCES A(AID) ON UPDATE PROPAGATE;

## Pergunta 18

Por responder

Pontuação 0,8

🚩 Destacar pergunta

Analyse the following SQL script:

```
pragma foreign_keys=ON;
```

```
drop table if exists Product;
```

```
drop table if exists Package;
```

```
create table Product (id integer, price real);
```

```
create table Package (id integer, idProd integer, amount integer);
```

Assume you are asked to modify the script so that the attribute "idProd" is a foreign key that corresponds to the "id" attribute in the Product relation. Additionally, the "idProd" attribute should never have a NULL value and when its value is updated we want the "cascade" behaviour to be used. A database engineer was given this task and came up with the answers shown below (assume the rest of the script remains identical to the one shown above). Which of the answers is correct?

Selecione uma opção de resposta:

☐ a. I don't want to answer

☐ b.

```
CREATE TABLE Package (id INTEGER PRIMARY KEY, idProd INTEGER REFERENCES Product NOT NULL ON UPDATE CASCADE, amount INTEGER);
```

☐ c. CREATE TABLE Package (id integer PRIMARY KEY, idProd INTEGER REFERENCES Product(id) on UPDATE CASCADE NOT NULL, amount INTEGER);

☐ d. CREATE TABLE Package (id INTEGER, idProd INTEGER REFERENCES Product ON UPDATE CASCADE NOT NULL, amount INTEGER);

☐ e. CREATE TABLE Package (id INTEGER, idProd INTEGER REFERENCES Product NOT NULL ON UPDATE CASCADE, amount INTEGER);

## Pergunta 19

Por responder

Pontuação 0,8

🚩 Destacar pergunta

Which statement is wrong about the PRIMARY KEY constraint in SQL?

Selecione uma opção de resposta:

☐ a. A PRIMARY KEY uniquely identifies each record in a table

☐ b. Primary keys must contain UNIQUE values

☐ c. I don't want to answer

☐ d. Primary key can be made of multiple columns

☐ e. Primary key can be made of any single columns

## Pergunta 20

Por responder

Pontuação 0,8

🚩 Destacar pergunta

A video streaming service has its database related to films and actors. A film consists of one or more actors. Each film has an identification number, a title and a duration. Each actor has a name and a nationality. Each actor is present at least in one film inserted in the database. The role of the actor in the film is classified by a name and a description. Both film and an actor may have received one or more awards. An award is determined by its name and description.

Considering the following sentences:

I - Film, actor and award are classes

II - The multiplicity between film and actor is many to many

III - Role is a attribute

Which ones are correct given the video streaming problem?

Selecione uma opção de resposta:

☐ a. I and II

☐ b. I don't want to answer.

☐ c. Only I

☐ d. I, II and III

☐ e. II and III

# Pergunta 21

Por responder

Pontuação 0,8

Destacar pergunta

Consider the following relations:

| Car |             |
|-----|-------------|
| car | model price |
| 320 | 15000       |
| 340 | 17500       |
| 420 | 25000       |

| Color |  |
|-------|--|
| color |  |
| White |  |
| Black |  |
| Blue  |  |

| Interior  |      |
|-----------|------|
| interior  | type |
| Leather   |      |
| Nylon     |      |
| Polyester |      |

In this system, Cars can be sold with all available interior colors, interior materials and outside paint.

Which of the following queries represents all possible combinations of cars, with varying interior types, interior colors and outside paint, but the colors of the interior and the car are different?

Which of the following queries represents all possible combinations of cars, with varying interior types, interior colors and outside paint, but the colors of the interior and the car are different?

Selecione uma opção de resposta:

- ☐ a. I do not wish to answer
- ☐ b.  $\sigma_{\text{Interior.color} \neq \text{Car.color}} (\text{Car} \times \text{Color} \times \text{Interior} \times \text{Color})$
- ☐ c.  $\text{Car} \times \text{Color} \times \text{Interior} \times (\rho_{\text{interiorColor}(\text{InteriorColor})} \text{Color})$
- ☐ d.  $\sigma_{\text{Interior.color} \neq \text{color}} (\text{Car} \times \text{Color} \times \text{Interior} \times (\rho_{\text{interiorColor}(\text{InteriorColor})} \text{Color}))$
- ☐ e.  $\sigma_{\text{Interior.color} \neq \text{Car.color}} (\text{Car} \times \text{Color} \times \text{Interior} \times (\rho_{\text{interiorColor}(\text{InteriorColor})} \text{Color}))$

# Pergunta 22

Por responder

Pontuação 0,8

Destacar pergunta

What's the equivalent of the following SQL query in Relational Algebra:

SELECT a,b FROM T1 NATURAL JOIN (SELECT \* FROM T2 WHERE c='John');

Selecione uma opção de resposta:

- ☐ a.  $\pi_{a,b} (T1 \bowtie \sigma_{c='John'} (T2))$
- ☐ b.  $\sigma_{a,b} (T1 \bowtie \pi_{c='John'} (T2))$
- ☐ c. I do not wish to answer
- ☐ d.  $\pi_{c='John'} (T1 \bowtie \sigma_{a,b} (T2))$
- ☐ e.  $\sigma_{a,b,c} (T1 \bowtie T2)$

## Pergunta 23

Por responder

Pontuação 0,8

Destacar pergunta

If two relations R and S are joined, then the non-matching tuples of both R and S are ignored in which type of join?  
(select correct option)

Selecione uma opção de resposta:

- ☐ a. I do not wish to answer
- ☐ b. Inner Join
- ☐ c. Full Outer Join
- ☐ d. Left Outer Join
- ☐ e. Right Inner Join

## Pergunta 24

Por responder

Pontuação 0,8

Destacar pergunta

Given R(A, B, C, D), S(A, C, E), what is the schema of  $R \bowtie S$  ?

Selecione uma opção de resposta:

- ☐ a. ACBDE
- ☐ b. ABCD
- ☐ c. I do not wish to answer
- ☐ d. ACE
- ☐ e. BDE

## Pergunta 25

Por responder

Pontuação 0,8

Destacar pergunta

Apply

| sID | cName    | major | dec |
|-----|----------|-------|-----|
| 12  | Stanford | CS    | Y   |
| 23  | MIT      | CS    | N   |
| 12  | MIT      | CS    | N   |



| sID | dec |
|-----|-----|
| 12  | Y   |
| 23  | N   |
| 12  | N   |

Selecione uma opção de resposta:

- ☐ a.  
 $\sigma_{(dec = 'Y' \vee dec = 'N')}$  Apply
- ☐ b. I do not wish to answer
- ☐ c.  
 $\pi_{(sID,dec)} (\sigma_{(sID = '12')})$  Apply
- ☐ d.  
 $\sigma_{(sID = '12' \vee sID = '23')}$  Apply
- ☐ e.  
 $\pi_{(sID,dec)}$  Apply